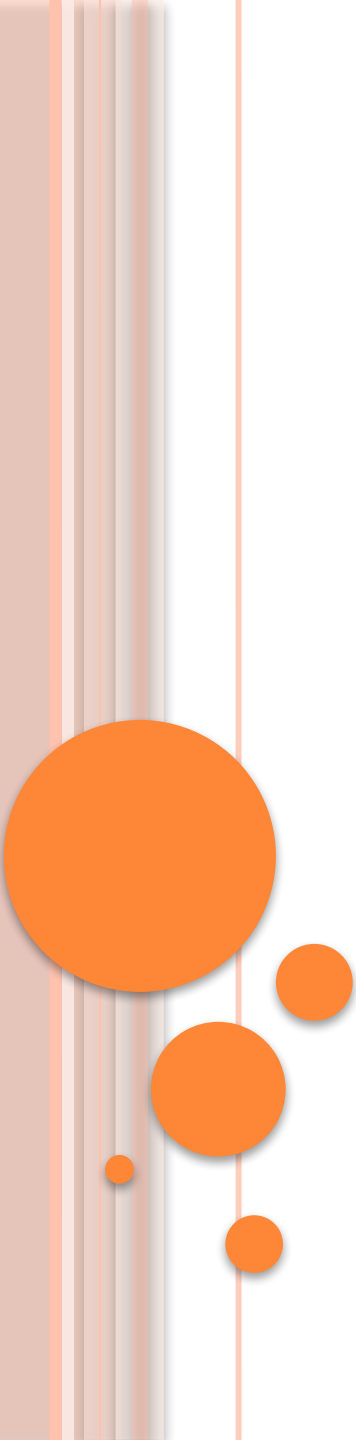




METHODOLOGY

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DEFINITION

- Schwardt (2007:195) defines research methodology as a theory of how an inquiry should proceed. It involves analysis of the assumptions, principles and procedures in a particular approach to inquiry. According to Creswell and Tashakkori (2007), and Teddlie and Tashakkori (2007), methodologies explicate and define the kinds of problems that are worth investigating; what constitutes a researchable problem; testable hypotheses; how to frame a problem in such a way that it can be investigated using particular designs and procedures; and how to select and develop appropriate means of collecting data.



METHODOLOGY

- It is the part of the proposal or research paper that describes the methods used to collect the data
- It follows the introduction and literature review
- It allows the reader to understand how the data were analyzed, and to judge for himself if the methods were justified
- It should be detailed enough for a good researcher to be able to replicate the study from reading the method section

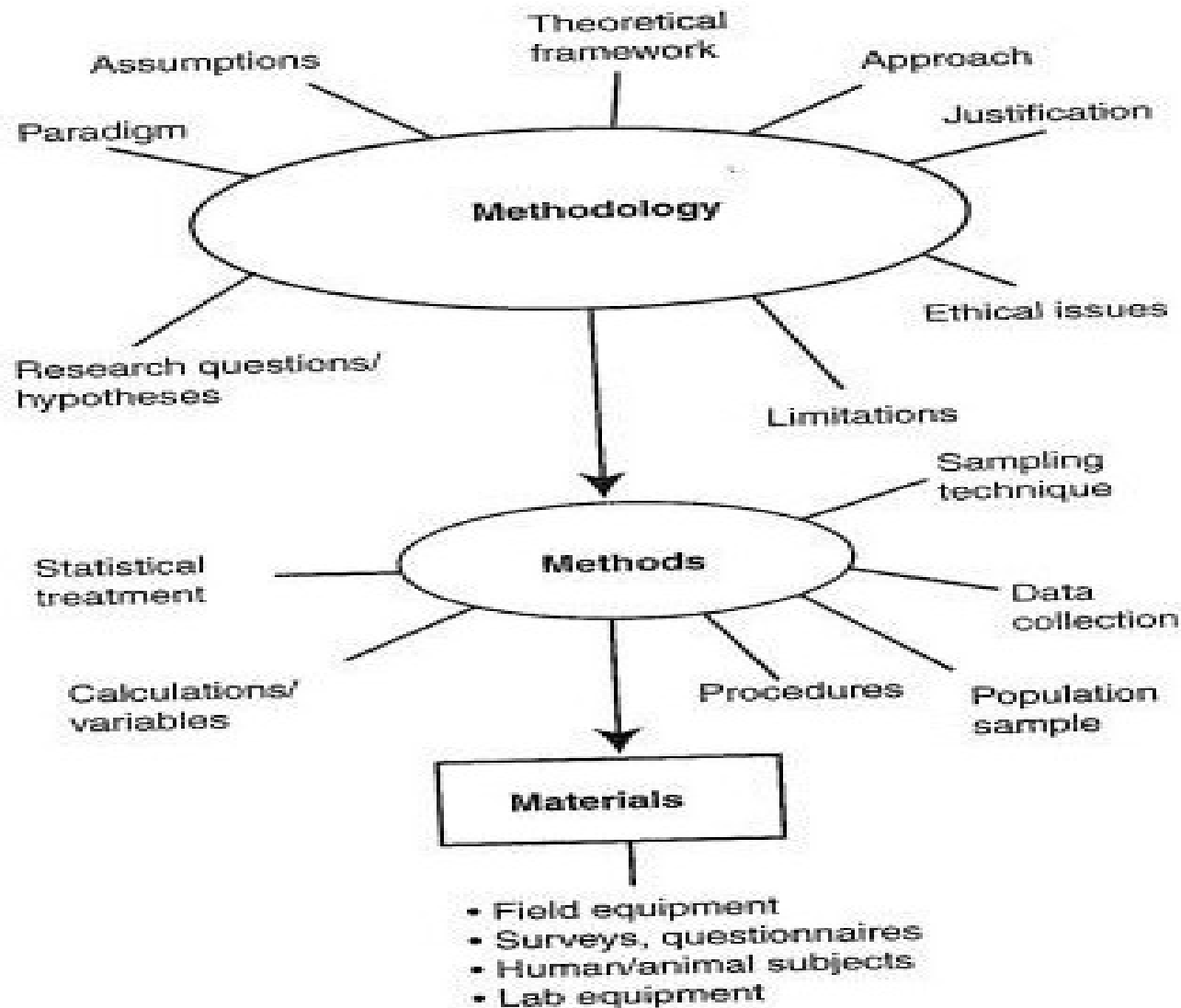


METHODOLOGY SECTIONS

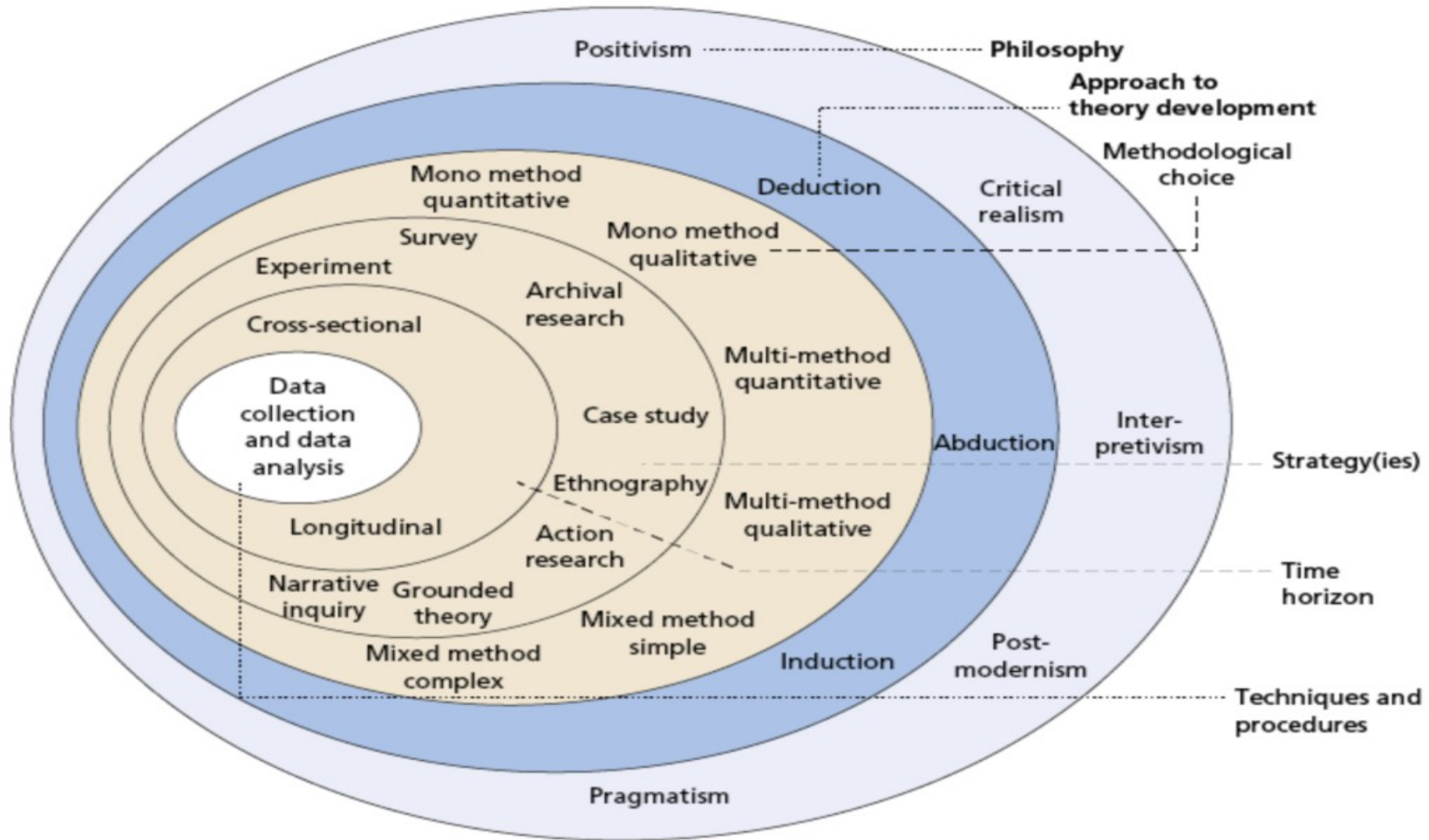
- Participants: Who was in the study
- Procedure: What happened in the study
- Measures/Materials: What measures were used—like surveys Or what materials—like special lab equipment
- Analysis section: Describes the statistical analysis



PROCESS



APPROACH



Source: ©2018 Mark Saunders, Philip Lewis and Adrian Thornhill

TO INCLUDE

- An outline of the research methods chosen (qualitative, quantitative, or mixed methods)
- Background and rationale for the methods chosen, explaining why one method was chosen over another
- Methods used for data collection and data analysis
- Materials and equipment used—keep this brief
- Difficulties encountered during data collection and analysis.
- A brief evaluation of your research explaining whether your results were conclusive and whether your choice of methodology was effective in practice



TO AVOID

- Irrelevant details, for example, an extensive review of methodologies (this belongs in the literature review section) or information that does not contribute to the readers' understanding of your chosen methods
- A description of basic procedures
- Excessive details about materials and equipment used. If an extremely long and detailed list is necessary, add it as an appendix
- Raw data



SUMMARY

Methodology	Approach
Method	Quantitative study
Paradigm	Positivism
Population	Household members of Dhaka city who are the potential user of IoT for security purposes
Selection of participants	Head of household
Sampling method	Snowball sampling
Unit of analysis	Individual
Data collection sources	Survey (primary data)
Research instruments	Adapted questionnaire
Likert scale	Seven
Time dimension	Cross-sectional analysis
Common method variance	Harman's single factor test
Data analysis technique	PLS-SEM
Data analysis tool	SPSS 23, SmartPLS 3.3.3
Sample size determination tool	G*Power 3.1
The minimum size of the sample	199
Number of accepted questionnaire	348



CHECKLIST

- **Introduce your methodological approach**, for example, quantitative, qualitative, or mixed methods.
- **Explain why your chosen approach is relevant** to the overall research design and how it links with your **thesis question**.
- **Justify your chosen method** and why it is more appropriate than others.
- **Provide background information** on methods that may be *unfamiliar* to readers of your thesis.
- **Introduce the tools that you will use for data collection**, and explain how you plan to use them (e.g., surveys, interviews, experiments, or existing data).
- **Explain how you will analyze your results**. The type of analysis used depends on the methods you chose. For example, exploring theoretical perspectives to support your explanation of observed behaviors in a qualitative study or using statistical analyses in a quantitative study.
- **Mention any research limitations**. Discussing the limitations justifies your choice of methodology despite the risks. It also explains under which conditions the results should be interpreted and shows that you have taken a holistic approach to your study.



FURTHER READING

- Sekaran, U. (2003). Research methods for business: A skill-building approach (4th ed.). New York: John Wiley & Sons.
- Bhattacharjee, A. (2012). Social science research: Principles, methods, and practices (2nd ed.). University of South Florida: Creative Commons Attribution. Retrieved from https://scholarcommons.usf.edu/cgi/viewcontent.cgi?article=1002&context=oa_textbooks
- Bryman, A. (2012). Social Research Methods (4th ed.), New York: Oxford University Press.
- Creswell, J. W. (2014). Research Design: Qualitative, Quantitative and Mixed Methods Approaches (4th ed.). Thousand Oaks, California: Sage.



THE END

Thank You

